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Superseding
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MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 49 MOD 0

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the MARK 49 MOD 0 impulse cartridge and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-46610	Primers, Percussion, Styphnate and Chlorate Types, for Small Arms Ammunition

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 1377

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General.

3.1.1 The MARK 49 MOD 0 impulse cartridge covered by this specification shall be manufactured in accordance with the drawings listed on LD 534525.

3.2 Materials. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referenced on the respective NAVAIR drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy cognizant field activity for cartridges prior to manufacture. When alternate materials or methods of cartridge manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 94 MARK 49 MOD 0 impulse cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purposes of this specification the primer (116M-282A) and the propellant charge are defined as primary components. See 4.3.3.

3.4 Primer. The primers shall be furnished from lots manufactured within the preceding twelve months. The primers shall be manufactured in accordance with Drawing B-6200959(116M-282A), loaded with Olin Industries primer mixture 282A; and shall be from lots which were accepted in accordance with MIL-P-46610, Class 1. See 4.7.1.

3.4.1 X-ray. Primers shall be free of the following defects when examined by x-ray in accordance with 4.7.2: missing or abnormally thin charge; charge broken, cracked or porous; cracked or off-center anvil or other misplaced or deformed inert parts; anvil abnormally close to primer cup dome; and all other observable defects in assembly.

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3.5 Propellant charges.

3.5.1 Quantity. All propellant charges shall be as specified on Drawing 2240805. Charges shall be determined by weight rather than by volume measurements.

3.6 Cartridge inspection.

3.6.1 Visual inspection. Cartridges shall be free of the following visible defects: dents, deep scratches, sharp edges, cracks, burrs, split or cracked lip in the crimped area, defective crimp of propellant can, incomplete or improper sealant application, cut or nicked "O" ring, or any other imperfections which could prevent entry into the firing chamber or adversely affect performance or resistance to moisture. See 4.7.3.

3.6.2 Leakage. Cartridge leakage rate shall not exceed 10^{-5} cc of air or gas per second when tested in accordance with 4.7.4.

3.6.3 X-ray. Cartridges shall be free of imperfections when examined by x-ray in accordance with 4.7.5.

3.7 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.8.1, shall meet the requirements listed below when subjected to the test prescribed in 4.8.1.

3.7.1 Chamber pressure. The upper (U) and lower (L) limits of maximum pressure shall be 900 psi and 450 psi, respectively.

3.7.2 Misfire. There shall be no misfires. See 4.8.2.1.

3.7.3 Hangfire. There shall be no hangfires. See 4.8.2.2.

3.8 Packaging, packing and marking. The packaging and packing of cartridges and the container markings shall be in accordance with Section 5. See 4.6.1.

3.9 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be directed to dimensions, finishes and coatings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-414 or MIL-STD-105, whichever applies.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-105 or MIL-STD-414, whichever applies.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 94 MARK 49 MOD 0 impulse cartridges shall be prepared. Of these cartridges, 84 shall be expended in the tests listed below and 10 shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.3 through 3.6.3 of this specification. Preproduction sample acceptance shall consist of the following tests as prescribed in MIL-D-21625: 40-foot drop (6), 15 "G" shock (12), temperature and humidity cycling (18), and vibration (12), Numbers in parentheses denote the number of cartridges allocated for test. These cartridges (with the exception of the 40-foot drop cartridges), after exposure to environmental treatment, together with 12 untreated cartridges shall be conditioned for a minimum of six hours at $+70^{\circ} \pm 5^{\circ}F$ and fired in the test fixture of 4.8.1. The cartridges shall be fired within five minutes after removal from conditioning. In addition, 24 cartridges shall be temperature conditioned for a minimum of six hours, 12 at $-65^{\circ} \pm 5^{\circ}F$ and 12 at $+160^{\circ} \pm 5^{\circ}F$, and fired in the test fixture of 4.8.1. These cartridges shall be fired within three minutes after removal from conditioning. All cartridges shall fire and produce results

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within the limits set forth in 3.7.1 and 3.7.3. The failure of any cartridge to meet the acceptance requirements as stated above shall cause rejection of the preproduction sample.

4.3.3 Production lot. Unless otherwise specified in the contract, the minimum and maximum production lot size shall be 181 and 500 cartridges, respectively. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.4 Gages. The contractor shall provide gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Quality conformance. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of LD 534525, and verification of the following requirements and tests:

Item	Nature of Requirement Verification	Applicable Paragraphs	Classification (WR-43)
Primary component	Single lot	3.3, 4.3.3	(M101)
Primer	Age, loading and acceptance	3.4, 4.7.1	(C1)
Primer	X-ray	3.4.1, 4.7.2	(C2)
Propellant	Type and weight	3.5.1	(C3)
Cartridge	Visual inspection	3.6.1, 4.7.3	(M102)
Cartridge	Leakage	3.6.2, 4.7.4	(C4)
Cartridge	X-ray	3.6.3, 4.7.5	(C5)
Cartridge	Ballistic test	3.7, 4.8	(C6)
Cartridge	Packaging, packing and container markings	3.8, 4.6.1	(M103)

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4.6.1 Packaging, packing and marking. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in Section 5.

4.7 Tests.

4.7.1 Primer, verification. The contractor shall provide verification that the primers used met all the requirements of 3.4.

4.7.2 Primer x-ray. Examine all primers by x-ray and reject those having any of the defects of 3.4.1. For the most revealing exposure, the primers shall be laid on their sides with the anvil cutouts vertical. Pressure sensitive tape may be used as an aid in positioning the primers.

4.7.3 Visual inspection. Examine all completely assembled cartridges and reject those having any of the defects noted in 3.6.1.

4.7.4 Leakage. Each cartridge in the preproduction sample or production lot sample shall be leak tested in a dry gas or dry air leak tester. Cartridges which exhibit a leak rate in excess of 10^{-5} cc/sec of air or gas at a pressure differential of 1.0 $\pm$ 0.1 atmosphere shall be considered defective. Cartridges of the preproduction sample which exhibit a leakage defect shall be rejected. If one or more cartridges of the production lot sample exhibit a leakage defect, then a 100 percent leak test of the lot being tested shall be required at no additional expense to the Government. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected.

4.7.5 X-ray examination. Examine all cartridges by x-ray and any observable imperfections in assembly shall be cause for cartridge rejection. Particular attention shall be given to the presence of the propellant charge.

4.8 Ballistic tests.

4.8.1 The number of cartridges selected from each production lot for the ballistic tests shall be in accordance with the requirements of Table A-2 of MIL-STD-414, Inspection Level IV and an AQL of 0.04. The cartridges shall be conditioned for a minimum of six hours at +70° $\pm$ 5°F and fired in a closed test chamber having an internal volume of 10.00 $\pm$ 0.15cc, excluding the volume of the cartridge, and a length to diameter ratio of approximately 0.73. The cartridges shall be fired within five minutes after removal from conditioning. A record of pressure versus time shall be taken by suitable instrumentation. The firing chamber shall be thoroughly cleaned and be free from residue or cartridge fragments before each firing. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from each production lot for investigative purposes.

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4.8.2 Criteria of acceptability. The upper (U) and lower (L) limits of maximum pressure shall meet the requirements of 3.7.1. Lot acceptability shall be determined in accordance with the applicable provisions of MIL-STD-414 at an AQL of 0.04.

4.8.2.1 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.8.2.2 Hangfire. One or more hangfires shall result in rejection of the lot represented. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and the start of pressure rise exceeds 0.05 second.

4.9 Retest. There shall be no retests.

4.9.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and that part of the test reconducted.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four cartridges shall be packaged in an inner container in accordance with the drawings listed on LD 534526.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be packaged in accordance with Method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 171-178.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Six inner containers as described in 5.1.1 shall be packed in accordance with the drawings listed on LD 534527.

5.2.2 Level B. Not applicable.

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5.2.3 Level C. When directed by the procuring activity, cartridges intended for expenditure in preproduction sample testing and for production lot acceptance tests and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3.2 Normal markings. Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on the drawings listed on LD 534526 and LD 534527, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 Intended use. The MARK 49 MOD 0 impulse cartridge was originally designed for use in the Seat Static Line Ballistic Cutter (Ordnance Associates P/N C250-5) used in the A-5 series aircraft.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required and, if so, specify the test activity.

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order.

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6.3.1 Test report. Results of ballistic tests shall be forwarded to the Naval Air Systems Command (AIR-53222A) and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. For the purposes of this specification, the following definitions apply.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 Explosives safety precautions. Safety precautions for explosive-loaded items are contained in MIL-STD-1314.

6.5.1 Applicability. The loading, assembly and handling of the explosives, explosive-loaded subassemblies and the finished items covered by this specification involve hazardous operations and therefore require suitable explosives safety precautions. At plants of the Naval Air Systems Command, Department of the Navy, and other Government plants where the Government is responsible for safety, the precautions listed in MIL-STD-1314 are mandatory. These same minimum provisions are recommended for observance by a contractor or manufacturer that explosively loads or assembles the item covered in this specification in order to fulfill his own safety responsibility.

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